CHAPTER 8

The special senses

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The special senses are hearing, sight, smell, taste and balance; all use specialised sensory receptors to collect and transmit information to specific areas of the brain. Incoming nerve impulses from the ears, eyes, nose and mouth are integrated and coordinated within the brain, allowing perception of this information. Up to 80% of what we perceive comes from external sensory stimuli. The first sections of this chapter explore the special senses, while the later ones consider the effect of ageing and consequences of disorders affecting the ear and the eye.

Hearing and the ear

Learning outcomes

After studying this section, you should be able to:

- describe the structure of the outer, middle and inner parts of the ear
- explain the physiology of hearing.

The ear is the organ of hearing and is also involved in balance. It is supplied by the cochlear branch of the 8th cranial (vestibulocochlear) nerve, which responds to the vibrations generated by sound waves and transmits this information to the brain.

Structure

The ear is divided into three distinct parts (Fig. 8.1): the outer ear, middle ear (tympanic cavity) and inner ear. The middle and inner ear are encased within the petrous part of the temporal bone.

The outer ear collects sound waves and directs them to the middle ear, which in turn transmits them to the inner ear. The inner ear converts the sound waves into nerve impulses, which are transmitted to the hearing area (auditory cortex) in the brain.

Outer ear

This consists of the auricle (pinna) and the external acoustic meatus (auditory canal).

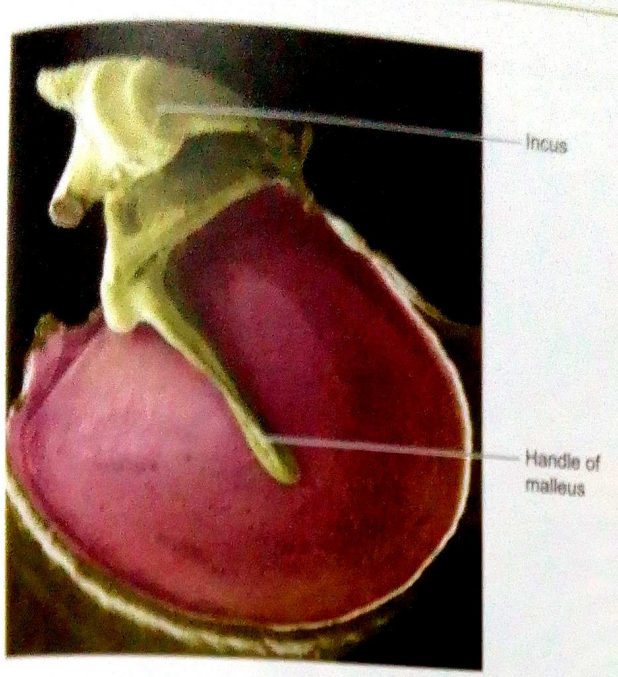


Figure 8.2 The tympanic membrane. Coloured scanning electron micrograph showing the malleus and the incus. (Steve G Schmeissner/Science Photo Library. Reproduced with permission.)

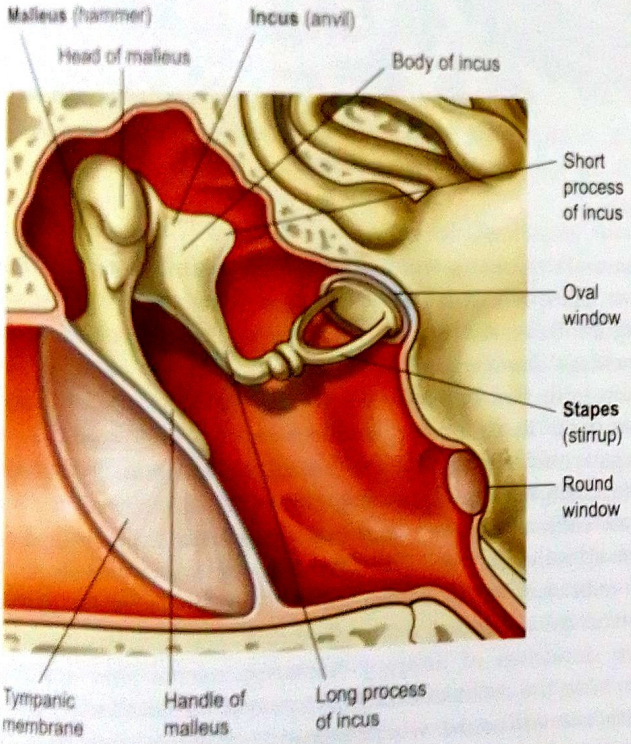


Figure 8.3 The auditory ossicles.

The medial wall is a thin layer of temporal bone in which there are two openings:

- oval window
- round window (see also Fig. 8.6).

The oval window is attached to one of the tiny bones in the middle ear, called the stapes and the round window by a fine sheet of fibrous tissue.

Air reaches the middle ear through the auditory (pharyngotympanic or Eustachian) tube, which links it to the nasopharynx. It is about 4 cm long and lined with ciliated columnar epithelium. The presence of air at atmospheric pressure on both sides of the tympanic membrane is maintained by the auditory tube and this enables the membrane to vibrate when sound waves strike it. The auditory tube is normally closed but when there is unequal pressure across the tympanic membrane, e.g. at high altitude, it is opened by swallowing or yawning and the ears 'pop', equalising the pressure again.

Auditory ossicles

These are three very small bones, only a few millimetres in size, that form a chain across the middle ear from the tympanic membrane to the oval window (Fig. 8.3; see also Fig. 8.1). Synovial joints between the ossicles allow them to vibrate within the air-filled tympanic cavity. The ossicles are held in place by fine ligaments and are named according to their shapes.

Malleus

This is the lateral hammer-shaped bone. The handle is in contact with the tympanic membrane, and the head forms a synovial joint with the incus.

Incus

This is the middle anvil-shaped bone. Its body articulates with the malleus, and the long process with the stapes; it is stabilised by the short process, fixed by fibrous tissue to the posterior wall of the tympanic cavity.

Stapes

This is the medial stirrup-shaped bone. Its head articulates with the incus and its footplate fits into the oval window.

Inner ear

The inner ear (Fig. 8.4) or labyrinth (meaning 'maze') contains the organs of hearing and balance. It has two structural parts: the bony labyrinth and the membranous labyrinth. It is divided into three main regions:

- the vestibule, containing the utricle and saccule
- three semicircular canals
- the cochlea.

The inner ear is formed from a network of channels and cavities in the temporal bone (the bony labyrinth). Within the bony labyrinth, like a tube within a tube, is the membranous labyrinth, a network of fluid-filled membranes that lines and fills the bony labyrinth (Fig. 8.4).

Bony labyrinth

This is lined with periosteum. Within the bony labyrinth lies the membranous labyrinth suspended in watery fluid called perilymph.

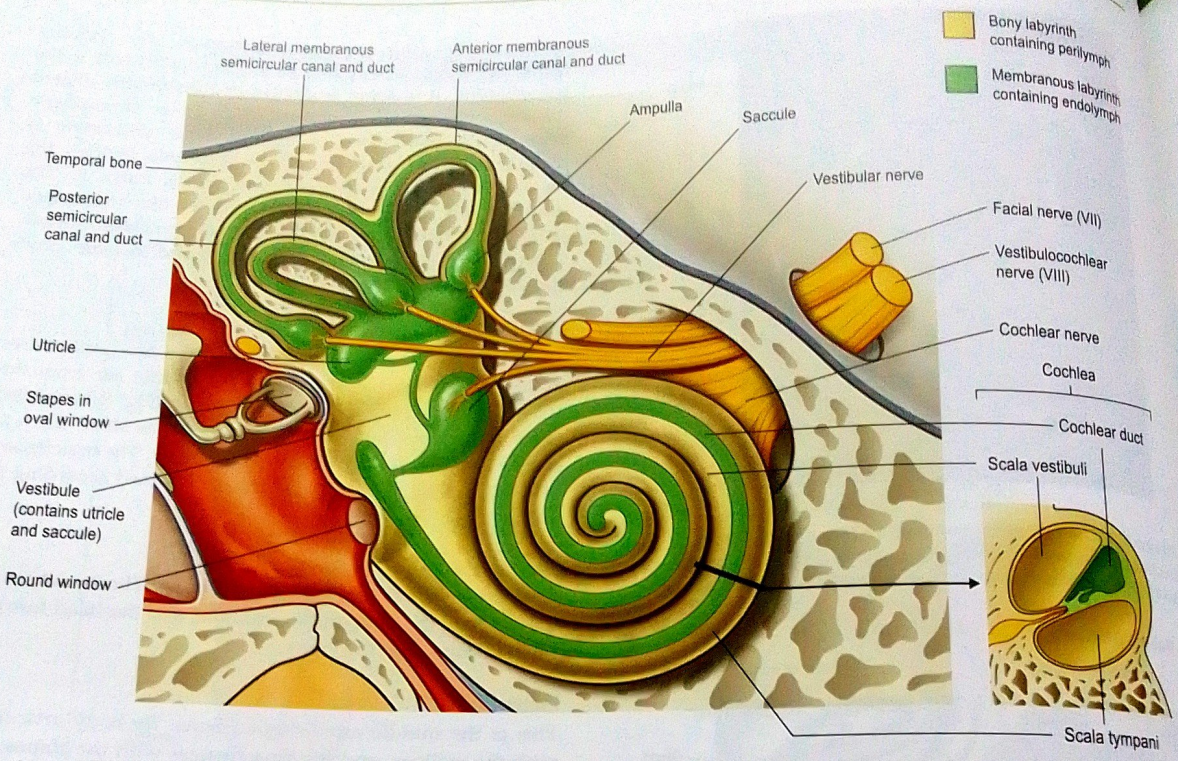


Figure 8.4 The inner ear. The membranous labyrinth within the bony labyrinth.

Membranous labyrinth
This is filled with endolymph.

Vestibule

This is the expanded region nearest the middle ear. The oval and round windows are located in its lateral wall. It contains two membranous sacs, the utricle and the saccule, which are important in balance (p. 218).

Semicircular canals

These are three tubes arranged so that one is situated in each of the three planes of space. They are continuous with the vestibule and are important in balance (p. 218).

Cochlea

This resembles a snail's shell in appearance. It has a broad base where it is continuous with the vestibule and a narrow apex, and it spirals round a central bony column.

A cross-section of the cochlea (Fig. 8.5) reveals its three compartments:

- the scala vestibuli
- the scala media, or cochlear duct
- the scala tympani.

Two of these compartments contain perilymph: the scala vestibuli, which originates at the oval window, and the scala tympani, which ends at the round window. These two compartments are continuous with each other and Figure 8.6A shows the relationship between them. The cochlear duct, which is an extension of the membranous labyrinth, lies between the scala vestibuli and the scala tympani, is triangular in cross-section and contains the organs of hearing. On the basilar membrane, the base of the triangle, are supporting cells and specialised cochlear hair cells containing auditory receptors. These cells form the spiral organ (of Corti), the sensory organ that responds to vibration by initiating nerve impulses which are then perceived as hearing within the brain. The auditory receptors are dendrites of afferent (sensory) nerves that combine, forming the cochlear (auditory) part of the vestibulocochlear (8th cranial) nerve, which passes through a foramen in the temporal bone to the hearing area in the temporal lobe of the cerebrum (see Fig. 7.20).

Physiology of hearing

Sound is carried as pressure (sound) waves in the air, which travel at about 340 metres per second. The shape of the auricle funnels and concentrates sound waves and directs them along the auditory canal, causing the tympanic membrane to vibrate.

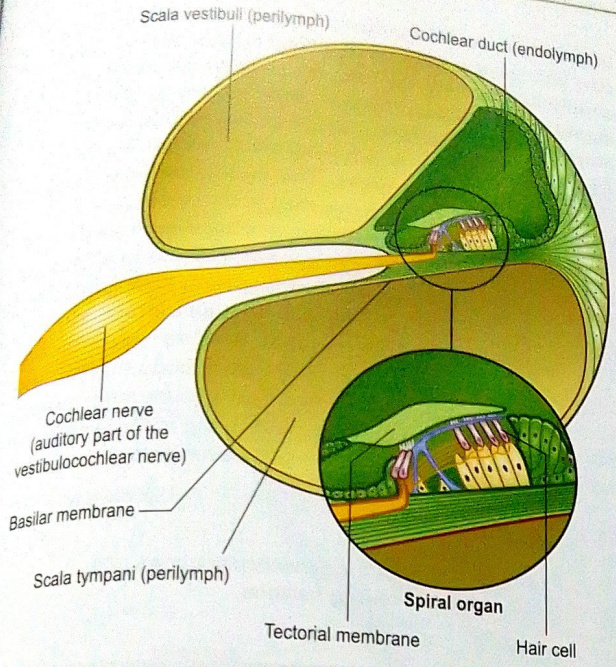


Figure 8.5 A cross-section of the cochlea showing the spiral organ (of Corti).

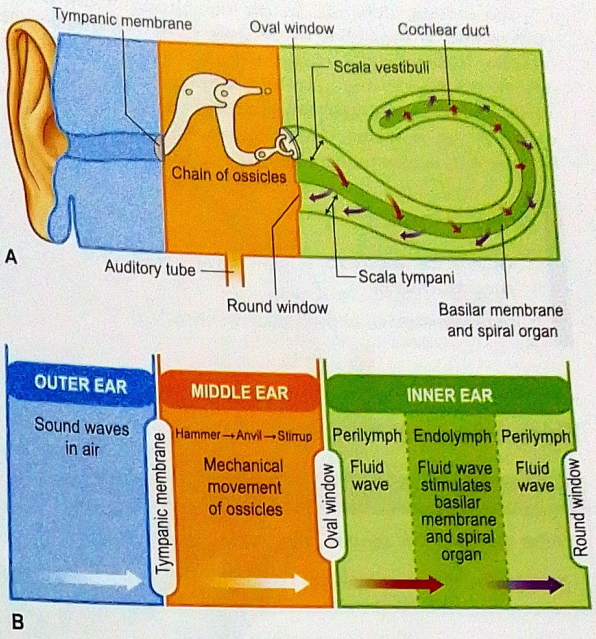


Figure 8.6 Passage of sound waves. (A) The ear with cochlea uncoiled. (B) Summary of transmission.

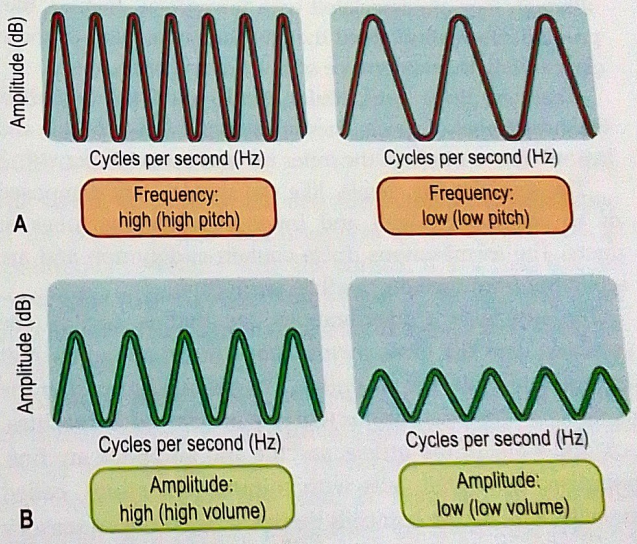


Figure 8.7 Behaviour of sound waves. (A) Difference in frequency but of the same amplitude. (B) Difference in amplitude but of the same frequency.

Tympanic membrane vibrations are transmitted and amplified through the middle ear by movement of the ossicles (Fig. 8.6).

At their medial end, the footplate of the stapes rocks to and fro in the oval window, setting up fluid waves in the perilymph of the scala vestibuli in the inner ear. Some of the force of these waves is transmitted along the length of the scala vestibuli and scala tympani, but most of the pressure is transmitted into the cochlear duct (Fig. 8.5). This causes a corresponding wave motion in the endolymph, resulting in vibration of the basilar membrane and stimulation of the auditory receptors in the hair cells of the spiral organ. The nerve impulses generated there are transmitted to the brain in the cochlear (auditory) branch of the vestibulocochlear nerve. The fluid wave is finally expended into the middle ear by vibration of the membrane of the round window.

The vestibulocochlear nerve transmits the impulses to the auditory nuclei in the medulla, where they synapse before they are conducted to the auditory area in the temporal lobe of the cerebrum (see Fig. 7.20). Because some fibres cross over in the medulla and others do not, the left and right auditory areas of the cerebrum receive impulses from both ears.

Sound waves have the properties of pitch and volume (Fig. 8.7). Pitch is determined by the frequency of the sound waves and is measured in hertz (Hz). Sounds of different frequencies stimulate the basilar membrane (see Fig. 8.6A) at different places along its length, allowing discrimination of pitch. The basilar membrane near the oval window is sensitive to high-pitched sounds, whereas the region towards the tip of the cochlea is responsive to lower-pitched sounds.

The volume depends on the magnitude of the sound waves and is measured in decibels (dB). The greater the amplitude of the wave created in the endolymph, the greater the stimulation of the auditory receptors in the hair cells in the spiral organ, and the louder the sound. Long-term exposure to excessive noise causes hearing impairment because it damages the sensitive hair cells of the spiral organ.

SPOT CHECK

1. Outline the features and functions of the external acoustic meatus.
2. Describe the role of the ossicles in hearing.

Balance and the ear

Learning outcome

After studying this section, you should be able to:

- describe the physiology of balance.

As well as its role in hearing, the ear is also the sense organ for balance, or equilibrium.

The semicircular canals and vestibule

The semicircular canals (see Fig. 8.4) have no auditory function, although they are associated with the cochlea. Instead, they provide information about the position of the head in space, contributing to maintenance of posture and balance.

There are three semicircular canals, one lying in each of the three planes of space. They are situated above, beside and behind the vestibule of the inner ear and open into it.

The semicircular canals, like the cochlea, are composed of an outer bony wall and inner membranous tubes or ducts. The membranous ducts contain endolymph and are separated from the bony wall by perilymph.

The utricle is a membranous sac that is part of the vestibule, and the three membranous ducts open into it at their dilated ends, the ampullae. The saccule is a part of the vestibule and communicates with the utricle and the cochlea.

In the walls of the utricle, saccule and ampullae are fine, specialised epithelial cells with minute projections, called hair cells. Among the hair cells there are receptors on sensory nerve endings, which combine, forming the vestibular branch of the vestibulocochlear nerve.

Physiology of balance

The semicircular canals and the vestibule (utricle and saccule) are concerned with balance, or equilibrium. The arrangement of the three semicircular canals, one in each plane, allows perception not only of the position of the head in space but also of the direction and rate of any movement. Any change of position of the head causes movement in the endolymph bathing the hair cells, which distorts them and stimulates the sensory receptors in the utricle, saccule and ampullae. The resultant nerve impulses are transmitted by the vestibular nerve, which joins the cochlear nerve to form the vestibulocochlear nerve. The vestibular branch passes first to the vestibular nucleus, then to the cerebellum.

The cerebellum also receives nerve impulses from the eyes and proprioceptors (sensory receptors) in the skeletal muscles and joints. The cerebellum coordinates incoming impulses from the vestibular nerve, the eyes and proprioceptors. Thereafter, impulses are transmitted to the cerebrum and skeletal muscles, enabling perception of body position and any adjustments needed to maintain posture and balance. This maintains upright posture and fixing of the eyes on the same point, independently of head movements.

SPOT CHECK

3. Outline the roles of the semicircular canals and vestibule in maintaining balance.

Sight and the eye

Learning outcomes

After studying this section, you should be able to:

- describe the gross structure of the eye
- describe the route taken by nerve impulses from the retina to the cerebrum
- explain how light entering the eye is focused on the retina
- state the functions of the extraocular eye muscles
- explain the functions of the accessory organs of the eye.

The eye is the organ of sight. It sits in the orbital cavity, a bony socket built into the bones of the face. The space between the eye and the orbital cavity is filled with adipose tissue. This together with the bony orbital cavity protects the eye from injury.

Structurally, the two eyes are separate but, unlike the ears, some of their activities are coordinated so that they normally function as a pair. It is possible to see with only one eye (monocular vision) but three-dimensional vision is impaired when only one eye is used, especially in relation to the judgement of speed and distance.

Structure

The eye is almost spherical in shape, about 2.5 cm in diameter and is supplied by the optic nerve (2nd cranial nerve).